1 Correct common denominator of 15 with unsimplified numerator $3(2x) - 5(x-1)$ **[M1]**

$\dfrac{x+5}{15}$ **[A1]**

[Total Marks: 2]

2 $\dfrac{5(3x)-4(x-2)}{20}$ or $\dfrac{15x}{20} - \dfrac{4x-8}{20}$ **[M1]**

$\dfrac{11x+8}{20}$ **[A1]**

[Total Marks: 2]

3 Uses common denominator $(x+2)(3x-1)$ **[B1]**

Correct unsimplified numerator $5(3x-1)+2(x+2)$ **[M1]**

$\dfrac{17x-1}{(x+2)(3x-1)}$ or $\dfrac{17x-1}{3x^2+5x-2}$ **[A1]**

[Total Marks: 3]

4 Common denominator of 12 and correct numerator formulation: $\dfrac{3(3x)-4(x-2)}{12}$ **[M1]**

Numerator expanded correctly to $9x - 4x + 8$ **[M1]**

$\dfrac{5x+8}{12}$ **[A1]**

[Total Marks: 3]

5 $\dfrac{5(2x)-3(x-4)}{15}$ or $\dfrac{10x}{15} - \dfrac{3(x-4)}{15}$ **[M1]**

$10x-3x+12$ or better as numerator (allow one sign error for M1, e.g., $10x-3x-12$) **[M1]**

$\dfrac{7x+12}{15}$ **[A1]**

[Total Marks: 3]

6 (a) Uses a common denominator of 12, e.g. $\dfrac{4(2x)+3(x-1)}{12}$ **[M1]**

$\dfrac{11x-3}{12}$ **[A1]**

(b) $3x^2$ or $\dfrac{1}{y^2}$ seen **[M1]**

$\dfrac{3x^2}{y^2}$ or $3x^2y^{-2}$ **[A1]**

[Total Marks: 4]

7 **(a)** $\dfrac{15x + 8x}{20}$ **[M1]**

$\dfrac{23x}{20}$ **[A1]**

(b) $\dfrac{5a^2b}{12} \times \dfrac{9}{10ab^2}$ **[M1]**

$\dfrac{3a}{8b}$ **[A1]**

[Total Marks: 4]

8 $(x - 4)(x + 4)$ seen **[B1]**

$(x - 4)(x - 1)$ seen **[B1]**

$\dfrac{x + 4}{x - 1}$ **[A1]**

[Total Marks: 3]

9 $3(x^2 - 4)$ or $3(x - 2)(x + 2)$ **[B1]**

$(x + 5)(x - 2)$ [B2] (or [M1] for $(x + a)(x + b)$ where $ab = -10$ or $a + b = 3$)

$\dfrac{3(x + 2)}{x + 5}$ or $\dfrac{3x + 6}{x + 5}$ **[A1]**

[Total Marks: 4]

10 **(a)** $\dfrac{3(x + 5) - 2(x - 2)}{(x - 2)(x + 5)}$ **[M1]**

$3x + 15 - 2x + 4$ correctly expanded on numerator **[M1]**

$\dfrac{x + 19}{(x - 2)(x + 5)}$ or $\dfrac{x + 19}{x^2 + 3x - 10}$ **[A1]**

(b) $(x - 4)(x + 4)$ seen for numerator **[M1]**

$(x - 4)(x + 2)$ seen for denominator **[M1]**

$\dfrac{x + 4}{x + 2}$ **[A1]**

[Total Marks: 6]

11 $(3x + 1)(x - 2)$ **[B2]**

If not B2, then B1 for $(3x + a)(x + b)$ where $ab = -2$ or $3b + a = -5$ **[B1]**

$(x - 2)(x + 2)$ **[B1]**

$(x - 2)(x + 1)$ **[B1]**

$\dfrac{3x^2 - 5x - 2}{x^2 - 4} \times \dfrac{x^2 - x - 2}{3x + 1}$ or equivalent correct inverted product **[M1]**

$\dfrac{(x - 2)(x + 1)}{x + 2}$ or $\dfrac{x^2 - x - 2}{x + 2}$ **[A1]**

[Total Marks: 6]

12 **(a)** Factorising numerator: $(2x+1)(x-3)$ **[B1]**

Factorising denominator: $(2x-3)(2x+3)$ **[B1]**

Note: no common factor cancels; the expression is already in its simplest form

Final answer: $\dfrac{2x^2-5x-3}{4x^2-9}$ or $\dfrac{(2x+1)(x-3)}{(2x-3)(2x+3)}$ **[A1]**

(b) Recognising $x^2-x-2=(x-2)(x+1)$ **[M1]**

Writing all over a common denominator of $(x-2)(x+1)$: $\dfrac{3(x+1)+4(x-2)-2x}{(x-2)(x+1)}$ **[M2]**

Simplifying numerator: $3x+3+4x-8-2x=5x-5$ **[M1]**

$\dfrac{5(x-1)}{(x-2)(x+1)}$ **[A1]**

[Total Marks: 7]

13 Factorising numerator to $(x-3)(x+3)$ **[B1]**

Factorising denominator to $(2x-1)(x+3)$ **[B1]**

$\dfrac{x-3}{2x-1}$ **[A1]**

[Total Marks: 3]

14 Numerator factorised as $(2x-3)(x-3)$ **[B2]**

Allow [M1] for $(2x+a)(x+b)$ where $ab=9$ or $2b+a=-9$

Denominator factorised as $2(x-3)(x+3)$ or $2(x^2-9)$ or $(2x-6)(x+3)$ **[B1]**

$\dfrac{2x-3}{2(x+3)}$ or $\dfrac{2x-3}{2x+6}$ **[A1]**

[Total Marks: 4]

15 $2(x-2)(x+2)$ **[B2]**

If not B2, then B1 for $2(x^2-4)$ or $(2x-4)(x+2)$ **[B1]**

$(x-2)(x+3)$ **[B2]**

If not B2, then B1 for $(x+a)(x+b)$ where $ab=-6$ or $a+b=1$ **[B1]**

$\dfrac{2(x+2)}{x+3}$ or $\dfrac{2x+4}{x+3}$ **[A1]**

[Total Marks: 5]

16 **(a)** Common denominator $(2x-1)(x+4)$ seen **[M1]**

Numerator: $(x+3)(x+4)-(x-2)(2x-1)$ **[M1]**

$(x^2 + 7x + 12) - (2x^2 - 5x + 2)$ correctly simplified to $-x^2 + 12x + 10$ **[A1]**

(b) Factorising numerator to $(x - 4)(x + 4)$ or denominator to $x(x + 4)$ **[M1]**

$\dfrac{x - 4}{x}$ **[A1]**

[Total Marks: 5]

17 Factorising $x^2 - 25$ to $(x - 5)(x + 5)$ and $x^2 - 3x$ to $x(x - 3)$ **[B1]**

Factorising $x^2 + 6x + 5$ to $(x + 1)(x + 5)$ and $2x - 6$ to $2(x - 3)$ **[B1]**

Showing multiplication by the reciprocal, e.g. $\dfrac{(x-5)(x+5)}{x(x-3)} \times \dfrac{2(x-3)}{(x+1)(x+5)}$ **[M1]**

$\dfrac{2(x-5)}{x(x+1)}$ (or equivalent expanded form $\dfrac{2x-10}{x^2+x}$) **[A1]**

[Total Marks: 4]

18 (a) $\dfrac{4(2x-1) - 3(x-3)}{(x-3)(2x-1)}$ **[M1]**

Numerator expanded correctly to $8x - 4 - 3x + 9$ **[M1]**

$\dfrac{5x+5}{(x-3)(2x-1)}$ or $\dfrac{5(x+1)}{(x-3)(2x-1)}$ **[A1]**

(b) Multiplies by reciprocal: $\times \dfrac{4x+2}{x+5}$ **[M1]**

Factorises $x^2 - 25$ to $(x - 5)(x + 5)$ or $4x + 2$ to $2(2x + 1)$ **[M1]**

Factorises denominator $2x^2 - 9x - 5$ to $(2x + 1)(x - 5)$ **[M1]**

2 **[A1]**

[Total Marks: 7]

19 (a) Uses common denominator $(x - 2)(x + 2)$ or $x^2 - 4$ **[B1]**

Correct unsimplified numerator $3(x + 2) - (x + 5)$ **[M1]**

$\dfrac{2x+1}{(x-2)(x+2)}$ or $\dfrac{2x+1}{x^2-4}$ **[A1]**

(b) Factorises $4x^2 - 25$ as $(2x - 5)(2x + 5)$ **[B1]**

Factorises denominators as $x(2x - 5)$ and $2x(2x + 5)$ **[B2]**

Allow [B1] if only one denominator is factorised correctly

Correct cancellation leading to final answer $3x$ **[A1]**

[Total Marks: 7]

20 (a) $\dfrac{5(2x+3) + 4(x-2)}{(x-2)(2x+3)}$ **[M1]**

$10x + 15 + 4x - 8$ **[M1]**

$\dfrac{14x+7}{(x-2)(2x+3)}$ or $\dfrac{7(2x+1)}{(x-2)(2x+3)}$ **[A1]**

(b) $\dfrac{2x^2-5x-12}{4x^2-9} \times \dfrac{2x-3}{x-4}$ **[M1]**

$(2x+3)(x-4)$ **[B1]**

$(2x-3)(2x+3)$ **[B1]**

1 **[A1]**

[Total Marks: 7]

21 (a) $(2x-1)(x+3)$ seen for numerator **[M1]**

$(2x-1)(2x+1)$ seen for denominator **[M1]**

Identifies and cancels the common factor $(2x-1)$ **[M1]**

$\dfrac{x+3}{2x+1}$ **[A1]**

(b) Factorises denominator: $x^2-4x+3=(x-1)(x-3)$ **[M1]**

Writes with a common denominator: $\dfrac{5(x-1)+(x+2)}{(x-1)(x-3)}$ **[M1]**

Simplifies numerator: $5x-5+x+2=6x-3$ **[M1]**

$\dfrac{3(2x-1)}{(x-1)(x-3)}$ or $\dfrac{6x-3}{x^2-4x+3}$ **[A1]**

[Total Marks: 8]